

Abhishek K M

Software Engineer — Backend / Full-Stack

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SUMMARY

Software Engineer (Backend / Full-Stack) — Final-year CS student with 3 production internships across deep tech, healthtech, and edtech. Shipped full-stack systems end-to-end — React/Next.js frontends, FastAPI/Node.js backends, AWS cloud infrastructure (EC2, RDS, SNS, VPC, Lambda), and LLM-powered AI/ML inference pipelines with real-time observability. Practiced agile/Scrum workflows, code review, and unit testing across all roles. Reduced API latency by 28% through LoRA fine-tuning and model quantization. Passionate about scalable system design, prompt engineering, and developer tooling.

WORK EXPERIENCE

BioCompute Inc Pvt. Ltd. · *Software Development Intern (concurrent with final semester)* Jan 2026 – Present

- Built and shipped production website and Admin CMS (Next.js, TypeScript, PostgreSQL, Supabase) with CI/CD via GitHub Actions — designed RESTful API endpoints, SMTP notification service for automated career listing events, and optimized animations for cross-device compatibility.
- Refactored a cross-platform Windows/Linux codebase into a unified Linux-native modular pipeline — eliminated environment-specific bugs, consolidated to a single shared codebase, and cut onboarding friction for new contributors through improved code review practices.
- Architected production-grade distributed system for edge deployment — microservices architecture with Docker, Nginx reverse proxy, Cloudflare DNS/SSL, Tailwind CSS UI, self-hosted on Raspberry Pi 5; achieved zero-downtime deployment at near-zero infra cost.

MyManah Pvt. Ltd. · *Software Development Intern* May 2025 – Aug 2025

- Built a full-stack mental health platform (React.js, FastAPI) with role-based auth and dynamic data pipelines; designed full system architecture on AWS — EC2, RDS PostgreSQL, SNS pub/sub, and VPC with gateway routing.
- Implemented serverless auth via AWS Lambda and Cognito; designed real-time AI risk-scoring feature routing requests through a FastAPI inference layer to quantized on-device LLM models.
- Optimized AI inference backend with LoRA fine-tuning and model quantization — reduced API latency by 28%; established CI/CD pipelines for automated deployments.

Crackverbal Edutech Pvt. Ltd. · *Tech Projects Intern* Jan 2025 – Mar 2025

- Engineered technical web infrastructure improvements — restructured HTML semantics to improve SEO score from 54 to 85, built event-tracked campaign landing pages with JavaScript analytics, and developed automated Python/Zapier pipelines integrating with Zoho CRM, reducing manual data-entry by ~3 hrs/week.

PROJECTS

RAG Observability Pipeline | [Live Demo](#) | *Next.js · FastAPI · ChromaDB · LangChain · HuggingFace · Groq LLaMA · Python*

- Built full-stack RAG pipeline with end-to-end observability — PDF ingestion, ChromaDB vector store (cosine similarity), Groq LLaMA generation; real-time observability dashboard surfaces per-query latency, token usage, and chunk confidence scores.
- Engineered Next.js dark-themed dashboard with chunk score visualizations, collapsible sidebar, post-ingest modal; implemented content fingerprint deduplication and split latency tracking across ingestion and retrieval microservices.

FinTrack | [Live Demo](#) | *Next.js · TypeScript · Drizzle ORM · Neon PostgreSQL · Chart.js · Vercel*

- Full-stack personal finance tracker with type-safe REST API, end-to-end data flow with Drizzle ORM, interactive Chart.js spending dashboards, and automated Vercel CI/CD deployment.

SKILLS

Languages: Python, JavaScript, TypeScript, C++, C

Frontend: React.js, Next.js, Tailwind CSS, HTML5/CSS

Backend: FastAPI, Node.js, Express.js, AWS Lambda, REST API design, microservices architecture, system design, distributed systems

Practices: Agile/Scrum, unit testing, code review, CI/CD, prompt engineering

Cloud & Infra: AWS (EC2, RDS, SNS, VPC, Lambda, Cognito), Docker, Nginx, Cloudflare, CI/CD (GitHub Actions), Kubernetes (familiar), Raspberry Pi edge

Databases: PostgreSQL, MySQL, MongoDB, Supabase, Neon, ChromaDB

AI/ML: LLM integration, RAG pipelines, LangChain, vector embeddings, LoRA fine-tuning, model quantization, prompt engineering, real-time inference optimization, observability

EDUCATION

Acharya Institute of Technology · Bachelor of Engineering (B.E.), Computer Science Nov 2022 – Nov 2026 · CGPA: 8.3

CERTIFICATIONS

AI For Everyone · DeepLearning.AI

Mar 2024 · ID: RGCA8ZMTMCY6

Agile with Atlassian Jira · Atlassian

Nov 2024 · ID: 53HJDIDALOYB

Introduction to Web Development with HTML, CSS, JavaScript · IBM

Aug 2024 · ID: 2XA0RKEFO5YK